

toring devices, household appliances and so on. It is estimated that M2M traffic will increase 24-fold between 2012 and 2017.

Taking e-health as an example, remote patient monitoring using a body area network, where a number of wireless sensors, both on-skin and implanted, record the patient's health parameters and send reports to a doctor, will soon become a reality. Therefore in order to offer e-health services, 5G will need to provide high bandwidth, meet stringent requirements for quality of service (such as ultra-low latency and loss-less video compression) and implement enhanced security mechanisms.

Work will be needed to manage radio resources efficiently, because of the diversity of traffic types, ranging from reports sent periodically by the meters to high-quality medical video transmission.

With new broadband services and high demand for mobile data, future wireless systems will require much higher capacity than can be provided today. There are three main ways of enhancing capacity, namely, dense deployment, additional spectrum bands and higher spectral efficiency.

Moving towards 5G will impose changes not only in the radio access network but also in the core network, where new approaches to network design are needed to provide connectivity to a growing number of users and devices. The trend is to de-couple hardware from software and move network functions towards software.

Conclusion

Despite never being able to successfully predict what each forthcoming generation of mobile technology would deliver in order to satisfy future users, the industry has, nonetheless, reached some consensus on the use cases for 5G. M2M communication is one. 5G should enable the IoT, the future where all our online-

enabled objects will quietly pass on data to our tech overlord of choice.

Facilitating the use of mobile networks by connected or autonomous cars, remotely-controlled industrial robots, tele-health systems and smartcity infrastructure are also all expected to figure large in 5G.

Industries and researchers are actively preparing for the growing communication needs of society, involving a combination of existing and evolving systems. 5G will comprise the set of technical components and systems needed to support these requirements and overcome the limits of current systems.

5G network addresses the world of the IoT. This changes the dynamics of wireless systems completely. It takes these from today's interference-limited environment, where interference from other mobiles radiates everywhere, like everyone yelling on the corner of a street, and now makes radio energy very focused like megaphones, with everyone talking with a megaphone to only whom they want.

We can imagine a world, possible by 2020, where almost anything that could be connected will be connected. Transition to 5G mobile communications is expected to include offloading traffic to unlicensed spectrum, improved carrier aggregation (up to 32 carriers), massive MIMO and support for a radio optimised for the low-end of the IoT market.

Before defining the road map to implement 5G networks, a lot of work has to be done ranging from testing the technology to defining performance requirements before commercial implementation.

Global 5G research is still taking place in isolation, but this will change. Various groups have different technology favourites, and standardisation discussions will start soon and technology selection will also begin in due course of time. **EFY**





PG08



PG09H

PG08 (Max. 10A)

- Power (W)
- Power Factor (PF)
- Current (mA)
- Voltage (V)
- Frequency (Hz)
- Total Usage Time (min)
- Total Used Energy (KWh)
- Carbon Emission (CO₂)

PG09H (Max. 15A)

- TRMS Voltage (V)
- TRMS Current (mA)
- Power (W)
- Energy Usage Time (EUT)
- Energy Consumption (KWh)

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